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Empowering sentiment analysis in social media: a comprehensive approach to enhance the classification of abusive Tamil comments using transformer models

Malliga Subramanian[1], Sathishkumar Veerappampalayam Easwaramoorthy[2], Nandhini Subbarayan[1] and Usha Moorthy[3*]

*Correspondence: m.usha@manipal.edu

[1] Department of Computer Science and Engineering, Kongu Engineering College, Perundurai, Erode, Tamil Nadu, India
[2] School of Computing and Artificial Intelligence, Faculty of Engineering and Technology, Sunway University, 47500 Petaling Jaya, Selangor Darul Ehsan, Malaysia
[3] School of Computer Engineering, Manipal Institute of Technology Bengaluru, Manipal Academy of Higher Education, Manipal, Karnataka, India

Abstract

Targeting individuals with abusive language based on social group membership such as gender, religion, or sexual orientation is a severe form of online verbal abuse. In today's digital world, the extensive spread of harmful and abusive content on social media has escalated, frequently fueling real-world violence against marginalized groups. Recognizing such content is especially difficult in code-mixed contexts and low-resource languages such as Tamil, where the linguistic complexity and data scarcity create substantial challenges. This study proposes a novel use of adapter-based tuning in multilingual transformer models, specifically mBERT, MuRIL (base and large), XLM-RoBERTa (base and large), and mDeBERTa, for classifying abusive comments in Tamil. We use the dataset provided by the DravidianLangTech@ACL 2022 shared task and increased the number of comments in training set from 2240 to 3742 using preprocessing and text augmentation methods. Unlike previous approaches that rely heavily on feature extraction and fine-tuning, this research shows that adapter-based models not only reduce the number of trainable parameters but also outperform fine-tuned and feature-extraction-based models in terms of classification accuracy, especially for imbalanced and under-resourced text data. Of the models explored, the adapter-based mDeBERTa achieved the highest accuracy of 75.393%, validating the effectiveness of parameter-efficient transfer learning for abusive language detection in low-resource, code-mixed contexts.

Keywords: Abusive comments, Transformer models, Adapter, Fine-tuned models, Multilingual models

Introduction

The internet facilitates rapid communication, connecting individuals worldwide. However, while many interactions are civil and constructive, abusive language often emerges, manifesting as disrespect, obscenity, and hostility, particularly against vulnerable groups such as women and children [1, 2]. Abusive language not only harms individuals but also negatively impacts communities and society at large. Abusive language is frequently used interchangeably with disrespectful language and hate speech [3, 4]. An increasing number of users have reported witnessing abusive behavior on social media in recent years. Such comments can be distressing for anyone, but they are especially hurtful when directed at children or teenagers who may lack the maturity or experience to cope with or understand them. In recent years, more people have reported abusive behavior on social media and these negative comments can undermine the confidence, potentially leading to emotions of despair, dissatisfaction, or even isolation from others.

Social media giants have used a variety of methods, such as human annotators, user reporting systems etc. to get rid of these kinds of comments and texts from their sites. Nevertheless, despite numerous attempts, the problem has not yet been solved. The main obstacles to identifying abusive language are its subjective perception and dependence on context [5]. Building a big, trustworthy dataset is challenging since even human annotators have trouble identifying abusive language. Recently, individuals from multilingual communities have begun posting and commenting using a mix of languages. The vocabulary and syntax of a phrase in the code-mixed format are derived from many languages [6]. Code-mixed data in Natural Language Processing (NLP) refers to data that contains a mixture of two or more languages. When multilingual people are unable to properly communicate in a single language, they frequently employ code-mixed text to convey themselves [7]. Annotating social media comments need automatic methods because it is impractical to manually identify such occurrences due to the large volume of content shared on social media. Owing to a lack of annotated datasets and other constraints, abusive comment identification in low-resource languages has mostly been disregarded [8]. Numerous researchers in the domain of NLP have been actively engaged in developing systems that aim to mitigate the dissemination of abusive content. These systems rely on machine learning models (Support Vector Machines, Naïve Bayes, K-Nearest Neighbor etc.), deep learning methods (Recurrent Neural Networks, GRU, LSTM etc.) and transformer models (BERT, mBERT, RoBERTa etc.).

Transformer models are widely used in many NLP applications including abusive comment detection because they process sequences in parallel, enabling faster training and inference compared to RNNs. They employ self-attention to identify the connections between all words in a sequence, allowing them to understand context clearly even in long or complex texts. Transformers can also take advantage of BERT and GPT, which can be adjusted for specific tasks using fewer computing resources. Their versatility across tasks, from text classification to machine translation, makes them a good choice in many NLP applications. The objective of this research is to employ transformer models for categorizing abusive content within a collection of Tamil posts taken from YouTube [9]. The use of transformer models for NLP applications in low resource languages such as Tamil is driven by their ability to handle complex language structures, leverage multilingual capabilities, and achieve high performance across various tasks, making them a powerful choice for developing language technologies. Hence, transformer-based models for identifying and categorizing abusive texts in Tamil by comprehending context and meaning have recently attracted a lot of attention from researchers. Prominent types of these models include BERT [10], mDeBERTa [11], RoBERTa [12], XLMRoBERTa [13], MuRIL [14], and others. Their performance has been noteworthy in effectively categorizing code-mixed texts from various languages [15]. There are three primary approaches to exploring these transformer models: (1) feature extractor (2) fine-tuning and (3) integrating adapters and subsequent training. While using as feature extractor, the pre-trained models process input text to generate embeddings or feature vectors without further training on the task-specific dataset. These embeddings can then be fed into a separate classifier for the desired tasks. Fine-tuning involves completely retraining these models to handle a new task and requires altering a vast number of parameters. In contrast, adapters are compact, lightweight modules inserted between transformer layers [16–18]. During model adaptation to a new task, the original transformer layer weights remain unaltered, with only the adapter layer weights being updated. According to He et al. [19], adapter-based tuning works better than fine-tuning for multilingual and resource-constrained tasks. This study detects abusive Tamil YouTube comments using feature extraction, fine-tuned, and transformer-based models. This research work focuses on the following research questions:

> Research Question 1 (RQ1): How well can pre-trained transformer models with adapters identify and classify abusive content on social media?
>
> For addressing this query, we undertook the process of fine-tuning and incorporating adapters into a range of transformer models, subsequently evaluating their efficiency.
>
> Research Question 2 (RQ2): How do text augmentation methods influence the performance of classification models?

To delve into this question, we carried out an ablation analysis that encompassed both augmented and non-augmented text scenarios. Subsequently, we assessed the effectiveness of various classifiers under these conditions. The key contributions of this study include the following:

i. We undertake the task of fine-tuning pretrained transformer models like mBERT, mDeBERTa, MuRIL, and XLMRoBERTa.
ii. We integrate a lightweight, small adapters into the transformer models and this integration was then assessed for its effectiveness in classifying abusive language in Tamil.
iii. Through the utilization of Bayesian Optimization, we adjust the hyperparameters to identify the ideal settings for the models.
iv. To address the issue of imbalanced datasets, we employ text augmentation using NLPAug. Furthermore, we conducted a comprehensive study that compares results with and without text augmentation.

To the best of our knowledge, this is the first study to address the gap in the literature on abusive comment detection by incorporating adapter models into transformer structures. Traditional approaches rely heavily on fine-tuning transformer models, which require retraining weights of all models, resulting in high computational costs and increased memory requirements, especially for large datasets. Our novel approach leverages adapter modules, which introduce extra, lightweight layers into the transformer model while leaving pre-trained model weights unchanged. This integration achieves two benefits namely efficiency in Parameter Management and Enhanced Task-Specific Optimization. By utilizing adapter modules, the overall number of parameters remains low compared to traditional fine-tuning approaches, which is crucial for resource-constrained environments like mobile or edge devices. This combination of methodologies tailored for the task potentially leads to improved performance and generalizability of the models, making it best suited for addressing the problem of abusive language classification in Tamil.

The remaining sections of the article are structured as follows: "Literature survey" section delves into the exploration of abusive language detection using a variety of models. In "Materials and methods" section, the dataset is summarized, and the data preparation for the study are elucidated. The developed models are explained in "Classifiers" section. "Experimental setup" section discusses the specifics of the experiments and training procedure. A review of the findings and their implications is given in "Empirical results and findings" section. The work is summed up in "Conclusion and future work" section, which also suggests future research areas.

Literature survey

The advent of transformers and pretrained language models has greatly influenced the methods used for identifying abusive language, with a strong emphasis on deep learning techniques. Additionally, there has been a push towards addressing abusive language in low-resource languages, leading to the publication of shared tasks that guide researchers in this direction. Below, we present an overview of the recent research attempts.

Shared tasks

Shared tasks are collaborative initiatives where researchers and practitioners work to address a common problem using a provided dataset. Typically organized as part of workshops or conferences, these tasks aim to advance the state of the art in a specific area by fostering the development and comparison of various approaches. The shared task [9] focused to detect and classify abusive language in Tamil, Malayalam, and Kannada, with the goal of encouraging the development of models to detect abusive content from social media. The analysis of the models submitted for this task was documented in a report by [20], outlining the dataset, methodologies, and outcomes of the models. Another shared task [21] involved the classification of abusive comments into eight categories, utilizing machine learning and transformer models, with an evaluation by [22] determining the MuRIL as the leading performer among others. Chakravarthi et al. [6] curated social media comments in low resource languages and presented them as a shared task and the findings were also presented. Additionally, Chakravarthi et al. [23] created a shared task [24] that aims to detect abusive contents from social media posts. Moreover, it was found that the most effective method for addressing data imbalance and multilingualism used XLM RoBERTa language model. Another shared task [25] has been introduced to detect hate speech and objectionable content in both English and Indo-Aryan languages. A study by [26] presented a comprehensive analysis of this shared task. These shared tasks aim to motivate scholars to tackle and make progress in issues related to the recognition of abusive text and to emphasize the necessity for further research.

Current Deep Learning Models

Given the dependance of machine learning models on feature extraction strategies, there has been a growing adoption of automated feature extraction methods. Furthermore, these automated techniques incorporate text representation and deep learning methodologies to boost the identification of abusive comments and improve overall performance. An overview of these models is presented below.

Ashraf et al. [27] examined identifying offensive comments on YouTube and tested various machine learning models (MLP, AdaBoost, RF, NB, SVM, LR, DT), along with Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) neural network models. AdaBoost achieved an F1-score of 91.96%, and CNN reached 91.68%. Lee et al. [28] explored hate speech detection on Twitter, comparing neural network models (CNN, RNN, GRU) with traditional machine learning methods. The bidirectional GRU achieved the highest F1-score of 80.5%. Emon et al. [29] evaluated Linear SVC, LR, MNB, RF, ANN, and RNN with LSTM for abusive Bengali text identification, with the LSTM-based RNN achieving an accuracy of 82.20%. In another study [30], 44,001 Facebook comments were analyzed using transformer models like as BERT and ELECTRA. The accuracy of ELECTRA was 84.9%, but BERT made it to 85%. A variety of machine learning models, such as LR, SVM, deep learning techniques like LSTM and BiLSTM with attention mechanism, and transformer models like m-BERT, Indic-BERT, and XLM-R, were provided by Sharif et al. [31] to detect offensive communication. Their findings indicated that XLM-R demonstrated superior performance in the context of Tamil and Malayalam, while m-BERT emerged as the top-performing model for detecting offensive comments written in Kannada.

Hande et al. [31] collected comments in Kannada from social media and categorized into hopeful speech or not-hopeful speech. This approach achieved a weighted F1-score of 75.6%, outperforming other models in comparison. Pitsilis et al. [32] suggested a strategy employing RNN classifiers to detect messages related to racism or sexism by incorporating user-specific information and word frequency vectors. The classifiers were evaluated on a 16 k-tweet corpus, demonstrating their improved ability to identify racism and sexism compared to existing algorithms. In another study [33], the identification of hope speech across English, Malayalam, and Tamil was explored using various machine learning techniques. It was shown that baseline algorithms perform well given sufficient training data. However, cross-lingual transfer learning using XLMRoBERTa emerged as the most effective algorithm in this context. Glazkova et al. [34] developed models to identify hate speech and offensive content and used Twitter-RoBERTa to detect the abusive comments, achieving F1-scores of 81.99% and 65.77% for the two subtasks. In a study by Steimel et al. [35], machine and deep learning models were proposed to classify English and German comments as abusive or non-abusive. Their findings suggested that achieving optimal performance in multilingual classification remains a challenging task, even when using comparable datasets.

El-Alami et al. [36] introduced an approach to classify offensive texts across multiple languages and this method leveraged transformer models such as BERT, mBERT, and AraBERT, specifically adapted for offensive comments detection in multilingual settings. The results indicated an increased accuracy and higher F1 scores. Sundar et al. [37] developed an approach to automatically detect abusive texts, employing cross-lingual word embeddings with language-independency. Empirical analysis compared this architecture with traditional, transformer, and transfer learning methods, yielding F1 scores of 61% for Tamil and 85% for Malayalam. In addition to model development, researchers have also established benchmark datasets in various initiatives [38–40]. These efforts also provide metrics for dataset evaluation and categorization, fostering future research attempts.

A thorough analysis of studies examining the application of several deep learning models for the goal of identifying hate speech in low-resource languages was presented by Sharma et al. [41]. They categorized abusive comments into several groups, including Homophobia, Xenophobia, Transphobia, Misandry, Misogyny, Counter-speech, and Hope speech, specifically analyzing Tamil and Tamil-English code-mixed content. The study utilized deep learning models such as RNN, LSTM and bidirectional LSTM for the classification task. Rehman et al. [42] proposed a method for learning social and text context elements with two different modules and these modules collaborate to form an integrated representation, which is then used to make the final prediction. The authors conducted extensive experiments on the SCIDN and MACI datasets to determine how well their method performed in comparison to other traditional and advanced methods. The SCIDN dataset comprises 1.5 million comments, whereas the MACI dataset has 665,000 multilingual remarks. The results showed that their solution outperform the existing top methods, improving the F1-score on the SCIDN and MACI datasets by 4.08% and 9.52%, respectively.

Chakravarthi et al. [43] presented a set of datasets containing abusive comments in Tamil and code-mixed Tamil-English, sourced from YouTube. To establish baseline performance on these datasets, experiments were conducted using various machine learning classifiers, with results reported based on F1-score, precision, and recall metrics. A t-test was also employed to see whether the classification results were statistically significant. The primary contribution of this work is the creation of a publicly available dataset of social media comments in Tamil, which contains classifications of abusive language in this under-resourced language. The findings indicate that MURIL was effective in binary classification of offensive remarks, highlighting the potential of multilingual transformers for this type of research. Sultana et al. [44] investigated the detection of abusive comments in Bengali on social media using machine learning. Six machine learning models such as LR, Multinomial NB, Random Forest, SVM, K-NN, and Gradient Boosting were tested, with SVM achieving the highest accuracy of 85.7%.

A review by Ibrohim et al. [45] surveyed abusive language detection studies using a systematic literature review, revealing that most rely on traditional machine learning and text representation approaches, primarily on Twitter data. It highlights challenges in enhancing Indonesian HSAL detection, including accurately identifying targets, categories, severity levels, and detecting spreaders like buzzers and fake accounts. Khan et al. [46] explored machine learning and deep learning models with attention mechanisms to identify abusive language in Urdu text. Text features were extracted using n-grams (Unigrams, Bigrams, Trigrams) through Count Vectorizer and TF/IDF. Four ML models such as LR, Gaussian NB, SVM, and Random Forest were evaluated, with Random Forest achieving the best results. Among DL models, the Bi-LSTM with attention and Word2Vec embeddings showed superior performance in detecting abusive language.

The study by Sharmila Devi et al. [47] proposes an automated system to identify and block code-mixed Tamil hate speech on platforms like Twitter and Helo. Hate speech intent is categorized into three classes: Targeted Individual (TI), Targeted Group (TG), and Others (O), each reflecting different types of abuse.

In summary, while there has been substantial research on detecting abusive comments using fine-tuned transformer models, one notable approach that remains unexplored is the use of adapter-based models. The existing works for classifying the abusive comment have not integrated adapter into transformer models. Also, they require more parameters to be retrained while finetuning them and taking more time for training. Adapter-based models function similarly to fine-tuning, except instead of changing the original pre-trained model, they add new layers and tweak their parameters while keeping the original model untouched. On the other hand, fine-tuning modifies the weights of the pre-trained layers themselves. Fine-tuning transformer models for specific tasks can be time-consuming and resource-intensive, making scaling difficult, particularly when working with huge datasets or real-time applications. Addressing these limitations may require exploring approaches that leverage adapter-based techniques to adapt transformer models to specific tasks more efficiently. In this work, we modify transformer models with adapters and assess the performance. The uniqueness lies in the utilization of adapters and tailoring the models to suit the specific dataset. Adding adapter modules improves the efficiency of parameters without losing performance. Additionally, we use Bayesian optimization to fine-tune a set of hyperparameters.

Materials and methods

Task description

This study used datasets from the Shared Task on Abusive Comment Detection in Tamil, presented as part of the DravidianLangTech workshop [21]. The datasets contain YouTube comments in Tamil and Tamil-English languages and do not contain any words in other languages. Due to the lack of existing datasets specifically for Tamil abusive comments, this research employed the mentioned dataset. However, by scraping data from other social media platforms, it is possible to create a more comprehensive dataset on Tamil abusive comments for future research.

Online abusive comments can take many forms such as insulting, intimidating, inciting violence or discrimination, targeting different groups or individuals based on gender, religion, sexual orientation, nationality, etc. Based on this, the comments in the dataset are categorized into one of the eight labels namely Misogyny, Misandry, Homophobia, Transphobia, Xenophobia, Counter Speech, Hope Speech and None-of-the-above. By categorizing comments into these specific classes, the dataset ensures a comprehensive approach to detecting and addressing various types of abusive texts.

The purpose of this task is to create methods capable of identifying abusive comments within a given set of Tamil texts. The dataset includes comments of varying sentence lengths, although the average sentence length in the corpus is typically one sentence. The training dataset consists of 2240 comments, with two comments not in Tamil being excluded from the analysis. Transformer models, such as BERT and GPT, have been pre-trained on extensive corpora and can subsequently be fine-tuned on smaller datasets for specific tasks. Since these models have already acquired a comprehensive understanding of language patterns during the pre-training, fine-tuning them with a training dataset of 2240 comments can still achieve good performance. The validation dataset contains 560 comments, while the test dataset comprises 699 comments. Each comment in both the training and validation datasets is labeled with one of the eight predefined categories. As part of the shared task itself, all the training, validation, and test datasets were given, hence we have not split the dataset into training, validation, or testing.

Preprocessing

The datasets employed in this study predominantly consist of words in Tamil, with a small proportion in English. Since these texts are collected from YouTube posts, they frequently include URLs, hashtags, and various special characters. To prepare the text for analysis, preprocessing is done, which includes eliminating irrelevant and extraneous characters. This helps to identify appropriate features that can facilitate the classification of the samples, and it encompasses the following steps:

Eliminating Emojis: The text messages contain emojis and emoticons within the dataset. The distribution of emojis across each class is as follows: None-of-the-above contains 390, Hope-Speech has 58, Misandry includes 43, Counter-speech has 17, Xenophobia contains 13, Misogyny has 9, while Homophobia and Transphobic both contain 0 emojis. These emojis can either be substituted with their corresponding textual representations or entirely removed. In this research, we employed their textual equivalents within the text. For instance, the clapping emoji has been replaced with the word "clapping" (கைதட்டல்) in Tamil. The 'demoji' package in Python is used to convert the emojis into texts. This involves converting emojis into text based on their common usage in social media rather than specifically aligning them with predefined classes in the dataset.

Removing Punctuation, Numbers, and Non-Tamil Text: In addition to replacing emojis and emoticons, we have eliminated excess white spaces, and punctuation marks such as ! and ?, as well as numerical digits. Although these characters may enhance readability, they do not contribute to discerning the underlying sentiment or stance in the text.

Text augmentation

Even though we can train the transformer models using small datasets, if the class labels are not balanced, they may be biased toward majority classes and poorly generalized to new, unseen data. To address this issue, we used text augmentation. We utilized the NLPAug Python library, which supports three distinct forms of text augmentation: character-level, word-level, and sentence-level augmentation, to facilitate text augmentation. In this study, we employed a variety of word- and sentence-level augmentation techniques, including synonym substitution, word replacement based on similar word embeddings, and context-based word replacement using robust transformer models. Because the "None-of-the-above" class is the most common category in the training dataset, we addressed the class imbalance by employing random undersampling. After the implementation of augmentation and undersampling techniques, the composition of the training dataset became more balanced. Table 1 shows the number of instances before and after applying augmentation and under-sampling techniques.

Table 1 Sample abusive comments on training dataset

Text	Label	Number of instances	
		Before Augmentation	After Augmentation
நியாயமுள்ள சில நல்ல மனிதர்களுள் நீங்களும் ஒருவர் ஐயா..ரொம்ப நன்றி ஐயா. You are one of the few good people who are fair sir..Thank you very much sir	Hope-speech	86	431
இருவரும் சேர்ந்து என்ன செய்தீர்கள்? What did you two do together?	Homophobia	35	421
உண்மை தான் என் கணவர் என் பேச்சைக் கேட்பது இல்லை ஆணவம் திமிர் இருக்கிறது மரியாதை கிடையாது பெண் என்றால் கேவலமா அவுங்க அம்மாவும் அக்காவும் பெண் தானே எவ்வளவு சுயநலம் மனது ரொம்பவும் வலிக்கிறது The truth is that my husband does not listen to me. He is arrogant and has no respect. His mother and sister are women. So Selfish. It is paining.	Misandry	447	447
ஆண்கள் தன் மனைவியை அதிகாரம் செய்வதாக நினைத்து ஒரு சில நேரங்களில் காயப்படுத்திகிறார்கள் Men sometimes hurt their wives by thinking they are empowering them	Counter-speech	149	448
ஊரில் உள்ள எல்லாப் பெண்களின் பேச்சையும் கேட்போம்.ஆனால் மனைவியின் பேச்சை மட்டும் கேட்க மாட்டோம்.அவள் நமது அடிமை. We will listen to all the women. But we will not listen to the wife alone. She is our slave.	Misogyny	126	409
கைபர் கணவாய் வழியாக வந்தீர்கள் நீ இந்தியன் இல்லை Your origin is not Indian and you journeyed through the Khyber Pass.	Xenophobia	95	416
நீ ஒரு பையன் இல்லை, நீ ஒரு அசிங்கமான திருநங்கை You're not a boy, you're an ugly transsexual	Transphobic	6	416
நல்ல மனிதர் ; வாழ்க நலமுடன். A good man; Live well	None of the above	1296	484
	Total	2240	2240

Classifiers

Till recently, RNNs and CNNs have been employed for constructing models in NLP-related tasks, including tasks like detecting offensive language, identifying sarcasm [48], performing sentiment analysis [49, 50], text summarization, and many more. However, these models face challenges in maintaining context of longer phrases, making it difficult to handle long-range dependencies. Additionally, their sequential nature limits the possibility of parallelization. To overcome these limitations, recent research has focused on utilizing transformer-based models, which pay attention to the current word being processed. The literature has highlighted two primary approaches for using transformer models: fine-tuning and adapter integration. In our study, we explore both techniques in addition to using them as word embedding techniques and provide a brief overview of the models below.

Classifiers with transformers

Transformers are designed to address extensive dependencies when tackling sequence-to-sequence tasks. The transformer model, introduced by Vaswani et al. [51], deviates from traditional recurrence-based models and it uses an attention system to find connections between the input and output across the entire data. This design promotes greater parallelization. Several research works have concentrated on harnessing the capabilities of transformers [52, 53]. Google's BERT takes advantage of bidirectional contextual comprehension from unannotated text by looking at both the words before and after a given word and this can handle variable-length sequences of tokens. However, BERT has a limit on how much text it can handle at once, usually up to 512 words. If the text is longer than this, it must be divided into smaller parts for BERT to process it. And any input sequences smaller than 512 tokens, will be padded with a special padding token called "[PAD]" to make it up to the maximum length of 512 tokens. This padding ensures that all input sequences fed into BERT are the same length. Recently, several variants of BERT have emerged, including ALBERT, RoBERTa, DeBERTa, XLMRoBERTa, mBERT, MuRIL, and more [54]. The maximum input length for different transformer models is determined by their architecture and tokenization process and this is 512 tokens for all the models used in this study. While optimizing input length could involve experimenting with smaller token lengths (e.g., 256 or 128 tokens), this often comes at the cost of increased time for hyperparameter tuning and additional experiments. The transformer-based NLP models were designed to utilize transfer learning [10, 54, 55]. Transfer learning involves initially training on extensive unlabeled text corpora [56] and subsequently fine-tuning the model for specific NLP tasks [10]. Given that labeled NLP datasets are often limited in size, training a model directly on such small datasets would result in suboptimal performance. As a result, pre-trained models are preferred in these situations. The next section will give an overview of four pre-trained conceptual frameworks serving as the foundation for the models used in this study.

mBERT

"Multilingual BERT," or mBERT [54], a self-learning model, has attracted a lot of attention due to its innovative approach to multilingual comprehension and its remarkable performance in a wide range of languages. One of the distinguishing features of mBERT is its training methodology. Instead of relying solely on monolingual English data, which is common in many NLP models, mBERT is trained on a vast and diverse multilingual dataset. This dataset consists of unlabeled raw text drawn from 104 languages, all sourced from Wikipedia. Furthermore, mBERT employs a vocabulary constructed from common word components, enabling it to work well with many languages without requiring large amounts of language-specific data.

XLMRoBERTa

RoBERTa, which stands for "A Robustly Optimized BERT Pretraining Approach," is a highly influential and advanced model in the realm of NLP. RoBERTa is primarily pre-trained on a large corpus of English text and it follows a similar pretraining strategy to BERT. A multilingual variant of RoBERTa is referred to as XLMRoBERTa [57], which was trained on a substantial 2.5 TB dataset sourced from CommonCrawl, encompassing content from 100 different languages. Similar to the RoBERTa model, XLMRoBERTa follows the same training methodology, eliminating the need for lang tensors to determine the language context. Notably, XLMRoBERTa does not rely on language-specific input markers or tokens to identify languages, making it a seamless choice for handling code-switching and mixed-language text.

MuRIL

MuRIL, a Multilingual Representations for Indian Languages [57], is a pioneering language model designed to address the unique linguistic diversity and challenges present in the context of Indian languages. MuRIL is built upon the transformer architecture, which is effective in capturing complex linguistic patterns and semantic representations. However, what distinguishes MuRIL is its focus on Indian languages, including but not limited to Hindi, Bengali, Tamil, Telugu, and more. A distinguishing feature of MuRIL is its ability to understand code-switching, a common linguistic phenomenon where people seamlessly switch between multiple languages or dialects in conversation. MuRIL's architecture and training process enables it to handle code-switching scenarios, making it well-suited for real-world language use cases.

mDeBERTa

DeBERTa, short for "Decoding-Enhanced BERT with Disentangled Attention" [11] represents an influential model in the NLP. Developed as an extension of the BERT family of models, DeBERTa introduces novel techniques to enhance the capabilities of language understanding and generation. The model employs disentangled attention, which allows it to focus on different aspects of the input text independently. Developed as an extension of DeBERTa, Multilingual DeBERTa is designed to handle text in multiple languages making it a flexible and useful model for many NLP tasks in different languages. All the above pretrained models are re-purposed in the following ways.

Feature extractor—word embedding

Here, context-based word representations (vectors) for text are produced using a transformer model that has already been trained. These word vectors are utilized as input characteristics to detect abusive comments detection. The pretrained model extracts the contextual embeddings from text comments and then feeds these embeddings into a custom classifier built on the top of pretrained models to determine the type of abusive comments. The upper layers in the pre-trained models are not modified to suit the dataset, but frozen with pre-trained weights.

Fine-tuning

Fine-tuning entails adjusting the weights of the pre-existing model and adapting them for a new task. This strategy reduces computational costs and allows us to build the most effective models. The pretrained models are then trained on a new classification problem as part of the fine-tuning phase, which also introduces new fully connected layers and then, a softmax layer called the classification layer is built up with as many neurons as there are classes in the dataset. During the fine-tuning process, various hyperparameters like the training epochs, learning rate, batch size etc. have been optimized. To acquire task-specific features, a selected number of upper layers in the pretrained models are unfrozen and retrained. In this experiment, fine-tuning and testing were performed on models such as mBERT, XLMRoBERTa (Base and Large), MuRIL (Base and Large), and mDeBERTa.

Adapter models

Recent research, as indicated by [58], demonstrates that fine-tuning typically achieves superior performance. However, the need to retrain and fine-tune a pre-trained model for each downstream task can result in training of huge parameters, leading to parameter inefficiency. To address this challenge, a parameter-efficient solution was proposed by Houslby et al. [18], which introduced bottleneck adapter modules. Adapter models in transformer architectures are a parameter-efficient way to adapt large models to new tasks without full retraining. They introduce small adapter layers between existing transformer layers, capturing task-specific information while leaving the main model layers fixed. This modular approach allows multiple adapters for different tasks or domains, making it ideal for multi-task learning and domain adaptation. Only the adapter layers are trained, significantly reducing training costs and storage needs while preserving the pre-trained knowledge. The adapter modules have exhibited significant performance in multi-tasking and cross-linguistic transfer learning, as discussed in [12, 59]. During retraining for the target task, the weights of all the pretrained models remain fixed, and only the weights of the integrated adapter modules are updated. This contributes to the effectiveness of adapter modules. The training process is outlined as follows:

- Introducing adapters to all transformer models
- Adding a softmax classification layer
- Setting the training parameters
- Retaining the weights of the original transformer model frozen
- Adjusting the task-specific parameters within the adapters

Adapter-based models have been proposed by recent attempts, including Houlsby et al. [18], Peters et al. [12], Pfeiffer et al. [60], and Kim et al. [61], to improve pre-trained transformer models. These studies often employed benchmark datasets covering tasks like sentiment analysis and question-answering. Nonetheless, the effectiveness of adapter modules for low-resource languages such as Tamil remains largely unexplored. In the present investigation, we leverage the adapter model introduced by Houlsby et al. [18] to classify abusive comments in Tamil, thus bridging this research gap. In our study, we repurpose each of the transformer models such as mBERT, mDeBERTa, MuRIL, and XLMRoBERTa independently for three distinct purposes: as feature extractors (word embedding), fine-tuners, and adapter-based models. Thus, we applied each transformer model individually for the above purposes, without combining their outputs in any ensemble fashion. The entire workflow of our approach is illustrated in Fig. 1. Our contribution is highlighted in this figure.

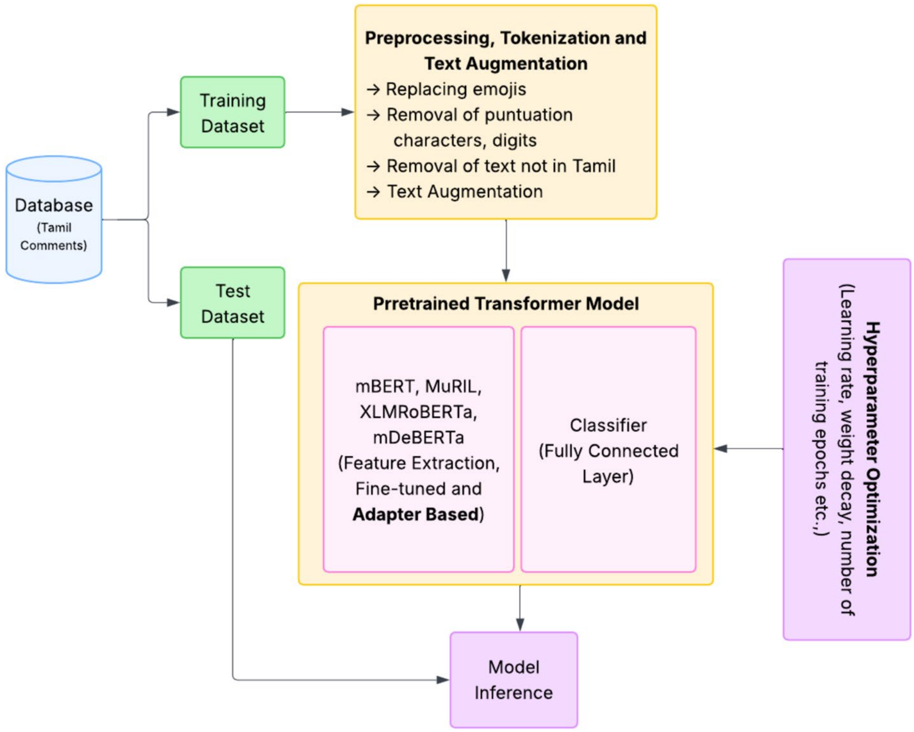

Fig. 1 Proposed workflow

Experimental setup

In this section, the experimental configuration, the training procedures for all pre-trained models, and a thorough performance analysis are discussed. We have conducted a range of tests to evaluate how well the developed models function as feature extractors, fine-tuners, and adapter-based transformers. The following section provides a comprehensive overview of these experiments.

Experimental platform

HuggingFace's Transformers library has been used for importing the pretrained models [62]. These models were extended with adapter-transformer, integrating adapter modules from AdapterHub. Due to the high computational demands of transformer-based models, they were run on a Graphical Processing Unit (GPU) for optimal performance. Textual augmentation was performed using the NLPAug Python library as part of the process.

Training and hyperparameter tuning

In this research, six transformer models were examined, including mBERT, mDeBERTa as well as XLMRoBERTa (Base and Large), along with MuRIL in both Base and Large versions. These transformers have been applied in three distinct approaches: for feature extraction, fine-tuning of model weights, and exclusive training of adapter modules.

Tuning of hyperparameters

This study has finetuned various hyperparameters and tuning helps find the hyperparameters that maximize the performance of the developed models on the training datasets, ensuring the models perform well on unseen data. The rationale behind finetuning the above hyperparameters are (1) tuning the number of epochs helps find a balance between adequate learning and computational efficiency, (2) proper tuning of the learning rate is critical for achieving the best model accuracy and ensuring stable convergence, (3) the choice of training batch size impacts the training speed and stability, (4) tuning of evaluation batch size helps balance between efficient computation and effective learning and (5) tuning the weight decay parameter ensures that the model can better generalize to a new data, improving its performance on validation and test sets. These hyperparameters have been optimized using an autonomous hyperparameter optimization software framework called Optuna, designed exclusively for machine learning. Optuna uses the Tree-structured Parzen Estimator (TPE) as its standard Bayesian optimization algorithm. Table 2 shows the set of hyperparameters that have been optimized in this work, together with their parameter space and the optimal values that were identified after several iterations.

Metrics for performance evaluation

Based on the accuracy, precision, recall, F1 score, macro, and weighted average, the developed models have been assessed. To compute these values, True Positive (TP), True Negative (TN), False Positive (FP), and True Negative (TP) have been calculated using Eqs. (1) to (4) for i from 1 to n classes.

$$tp_i = c_{ii} \text{ for 'i' from 1 to n} \tag{1}$$

Here tp_i represents true positive for ith class. For instance, tp1 represents the true positive for the "hope speech" class, that is $tp_i = c_{ii}$ means tp1 = c11. The false positive for class "hope speech" is sum of 1st column minus tp_{11} in a confusion matrix. To generalize the false positives for all classes, the above equation is defined. C_{ii} is not a matrix. It is used to represent C_{11}, C_{22}, C_{33} etc. (diagonal elements in the confusion matrix.)

$$fp_i = \sum_{l=1}^{n} (c_{li}) - tp_i \tag{2}$$

Table 2 Tuning of Hyperparameters

Hyperparameters	Parameter space	mBERT	MuRIL		mDeBERTa	XLM-RoBERT	
			Base	Large		Base	Large
Learning rate	0.01 to 0.0004	0.009	0.006	0.009	0.009	0.008	0.007
Number of Training Epochs	40 to 200	142	75	95	101	98	103
Weight decay	0.01 to 0.00004	0.000042	0.000064	0.00092	0.00082	0.00079	0.000091
Batch size for training	32,64,128, 256	64	64	128	64	32	64
Batch size for evaluation	32,64,128, 256	32	64	64	64	32	32

$$fn_i = \sum_{l=1}^{n} (c_{il}) - tp_i \tag{3}$$

$$tn_i = \sum_{l=1}^{n} \sum_{k=1}^{n} (c_{lk}) - tp_i - fp_i - fn_i \tag{4}$$

The accuracy, precision, recall and F1-score have been computed as follows:

$$Accuracy = ((TP + TN)/(TP + TN + FP + FN) \tag{5}$$

$$Recall = TP/(TP + FN) \tag{6}$$

$$Precision = TP/(TP + FP) \tag{7}$$

$$F1\ score = (2 * Precision * Recall)/(Precision + Recall) \tag{8}$$

$$Class\ Accuracy_i = (TP_i + TN_i)/Total\ Samples \tag{9}$$

As shown in Table 1, there is not an equal distribution of the instances of each class. The weighted average biased toward classes that occur more frequently may be more likely to underestimate the error in the class that does so. Hence, the macro average, which treats each class equally, can overcome this problem by averaging the precision and recall across all classes. Consequently, in infrequent cases, the performance of the model can be more accurately measured.

Empirical results and findings

The performance of the developed models is demonstrated in this section, along with a comparison of their results with those of other models that have been evaluated on the same dataset.

Feature extraction—results

During feature extraction, context-based word representations are generated from text comments using pre-trained models. After that, a classifier added on top of the previously trained models receives and classifies these representations. Table 3 presents the results of the classification.

Fine-tuned models—results

During the process of fine-tuning, the models have been retrained specifically for the dataset under consideration. To assess the performance of fine-tuned models, which have been trained on the training dataset, we conducted an evaluation using the test dataset to determine their performance on unseen data. The findings are presented in Table 4.

Based on the findings presented in Table 4, it can be inferred that mDeBERTa exhibits superior accuracy in comparison to the other models. mDeBERTa is developed to manage multilingual text data, empowering it to proficiently handle and interpret intricacies

Table 3 Performance of feature extraction

Classifiers	Class labels	Accuracy (%)	Recall (%)	Precision (%)	F1-Score (%)
mBERT	Hope-speech	51.502	23.077	22.222	22.642
	None-of-the-above		53.606	81.091	64.544
	Homophobia		25.000	16.667	20.000
	Misandry		65.354	43.915	52.532
	Counter-speech		46.809	21.359	29.333
	Misogyny		39.583	27.536	32.479
	Xenophobia		20.000	20.833	20.408
	Transphobic		0.000	0.000	0.000
	Macro/weighted average				
	Macro average		34.179	29.203	30.242
	Weighted average		51.502	61.329	53.961
XLMRoBERTa—base	Hope-speech	56.223	42.308	18.966	26.19
	None-of-the-above		58.413	84.375	69.034
	Homophobia		37.5	17.647	24
	Misandry		66.142	54.194	59.574
	Counter-speech		44.681	25.926	32.813
	Misogyny		39.583	31.148	34.862
	Xenophobia		48	31.579	38.095
	Transphobic		0.000	0.000	0.000
	Macro/weighted average				
	Macro average		42.078	32.979	35.571
	Weighted average		54.223	63.980	59.120
XLMRoBERTa—large	Hope-speech	60.944	61.538	29.630	40.000
	None-of-the-above		62.019	88.966	73.088
	Homophobia		50.000	30.769	38.095
	Misandry		70.866	59.603	64.748
	Counter-speech		57.447	29.670	39.130
	Misogyny		41.667	31.746	36.036
	Xenophobia		40.000	29.412	33.898
	Transphobic		50.000	33.333	0.000
	Macro/weighted average				
	Macro average		54.192	41.641	40.625
	Weighted average		60.944	62.552	63.503
MuRIL—base	Hope-speech	53.648	50.000	18.571	27.083
	None-of-the-above		52.644	87.600	65.766
	Homophobia		50.000	25.000	33.333
	Misandry		75.591	46.154	57.313
	Counter-speech		31.915	23.077	26.786
	Misogyny		39.583	26.389	31.667
	Xenophobia		36.000	50.000	41.860
	Transphobic		0.000	0.000	0.000
	Macro/weighted average				
	Macro average		41.967	34.599	35.476
	Weighted average		53.648	66.648	56.414

Table 3 (continued)

Classifiers	Class labels	Accuracy (%)	Recall (%)	Precision (%)	F1-Score (%)
MuRIL—large	Hope-speech	57.94	46.154	22.642	30.380
	None-of-the-above		62.981	86.469	72.879
	Homophobia		12.500	4.762	6.897
	Misandry		62.205	56.028	58.955
	Counter-speech		44.681	25.301	32.308
	Misogyny		41.667	33.898	37.383
	Xenophobia		40.000	26.316	31.746
	Transphobic		0.000	0.000	0.000
	Macro/weighted average				
	Macro average		38.773	31.927	33.818
	Weighted average		57.940	67.507	61.168
mDeBERTa	Hope-speech	54.936	42.308	16.418	23.656
	None-of-the-above		53.365	89.157	66.767
	Homophobia		62.500	23.810	34.483
	Misandry		77.953	48.529	59.819
	Counter-speech		36.170	25.373	29.825
	Misogyny		41.667	28.169	33.613
	Xenophobia		40.000	50.000	44.444
	Transphobic		0.000	0.000	0.000
	Macro/weighted average				
	Macro average		44.245	35.182	36.576
	Weighted average		54.936	68.189	57.781

within a spectrum of languages. Its resilient structure facilitates the assimilation of cross-lingual information, enhancing its accuracy in effectively classifying textual data from a diverse array of languages.

Adapter models—results

Additionally, we added adapter-based modules to the transformer because fine-tuning needs to retrain a lot of parameters. These modules were trained and tested, and they only needed a few parameters to be retrained. Like the fine-tuned models, we tested the adapter-based models that were customized to the training dataset to make sure they would classify an unseen dataset fairly. The findings are detailed in Table 5.

According to the data presented in Table 5, mDeBERTa has the maximum accuracy of 75.393%, whereas mBERT exhibits the lowest accuracy of 56.08% among the models under consideration. mDeBERTa utilizes advanced representation learning techniques, enabling it to grasp intricate contextual relationships within the textual data. This model fosters a deeper and refined comprehension of language, facilitating the accurate classification of diverse textual inputs. Figures 2 and 3 present the confusion matrices for the fine-tuned and adapter-based approaches, respectively, offering a visual interpretation of the performance. As illustrated in Fig. 4, the adapter-based mDeBERTa model demonstrates notably strong performance across most abusive comment classes.

Table 4 Performance of Finetuned models

Classifiers	Class labels	Accuracy (%)	Recall (%)	Precision (%)	F1-Score (%)
mBERT	Hope-speech	53.791	26.923	24.138	25.455
	None-of-the-above		55.048	83.273	66.281
	Homophobia		25.000	66.667	36.364
	Misandry		66.929	41.463	51.205
	Counter-speech		51.064	22.430	31.169
	Misogyny		45.833	37.288	41.121
	Xenophobia		28.000	33.333	30.435
	Transphobic		0.000	0.000	0.000
	Macro/weighted average				
	Macro average		37.350	38.574	35.254
	Weighted average		53.791	64.014	56.120
XLMRoBERTa—Base	Hope-speech	61.946	57.692	28.302	37.975
	None-of-the-above		61.779	88.316	72.702
	Homophobia		50.000	50.000	50.000
	Misandry		70.866	58.824	64.286
	Counter-speech		63.83	32.258	42.857
	Misogyny		50.000	36.364	42.105
	Xenophobia		52.000	38.235	44.068
	Transphobic		0.000	0.000	0.000
	Macro/weighted average				
	Macro average		50.771	41.537	44.249
	Weighted average		61.946	70.906	64.281
XLMRoBERTa—large	Hope-speech	63.09	69.231	36.735	48
	None-of-the-above		63.221	91.638	74.822
	Homophobia		62.5	50	55.556
	Misandry		74.803	60.51	66.901
	Counter-speech		65.957	33.696	44.604
	Misogyny		52.083	38.462	44.248
	Xenophobia		52.000	35.135	41.935
	Transphobic		50.000	50.000	0.000
	Macro/weighted average				
	Macro average		61.224	49.522	47.008
	Weighted average		64.521	73.776	66.643
MuRIL—base	Hope-speech	56.080	61.538	23.881	34.409
	None-of-the-above		54.327	87.597	67.062
	Homophobia		62.500	55.556	58.824
	Misandry		77.953	47.143	58.754
	Counter-speech		40.426	28.358	33.333
	Misogyny		43.75	28.767	34.711
	Xenophobia		36.000	60.000	45.000
	Transphobic		0.000	0.000	0.000
	Macro/weighted average				
	Macro average		47.062	41.413	41.512
	Weighted average		56.509	68.250	58.773

Table 4 (continued)

Classifiers	Class labels	Accuracy (%)	Recall (%)	Precision (%)	F1-Score (%)
MuRIL—large	Hope-speech	59.94	61.538	29.091	39.506
	None-of-the-above		66.106	89.869	76.177
	Homophobia		75	28.571	41.379
	Misandry		68.504	58.784	63.273
	Counter-speech		57.447	34.177	42.857
	Misogyny		47.917	42.593	45.098
	Xenophobia		56.000	41.176	47.458
	Transphobic		0.000	0.000	0.000
	Macro/weighted average				
	Macro average		60.314	46.783	44.469
	Weighted average		64.235	72.412	66.451
mDeBERTa	Hope-speech	66.715	69.231	31.034	42.857
	None-of-the-above		63.942	94.662	76.327
	Homophobia		75.000	54.545	63.158
	Misandry		86.066	58.659	69.767
	Counter-speech		59.574	38.356	46.667
	Misogyny		52.083	35.714	42.373
	Xenophobia		60.000	68.182	63.830
	Transphobic		0.000	0.000	0.000
	Macro/weighted average				
	Macro average		58.237	47.644	50.622
	Weighted average		66.715	76.370	68.741

In the test dataset, there are 48 comments classified as counter speech, 8 classified as homophobia, 26 as hope speech, 127 as misandry, 47 as misogyny, 416 as none of the above, 2 as transphobic, and 25 as xenophobic. Given that many instances in the test dataset fall under the "None of the above" class, Figs. 2, 3, 4 reflect a higher number of instances in this category.

Performance comparison

In this study, we performed a comparative analysis of adapter-based models with other existing classification models, using the same shared task dataset as the proposed work and evaluated the efficiency of the proposed models. For comparison, we have included not only transformer-based models but also classical baseline methods, such as SVM with TF-IDF and SMOTE [64], BiLSTM [69], Linear SVC [74], and others. These models serve as direct comparative baselines for evaluating the performance achieved by adapter-based transformer models. The results of the comparison are summarized in [63]. The same is also outlined in Table 6.

From the literature review of previous research using the same dataset, we found that some studies used SMOTE for text augmentation, while others used the NLPAug library and transliteration to balance the dataset. However, most of the works used the original imbalanced dataset and focused on F1-score (rather than accuracy) to

Table 5 Evaluation of adapter-based transformer models

Classifiers	Class Labels	Accuracy (%)	Recall (%)	Precision (%)	F1-Score (%)
mBERT	Hope-speech	56.08	42.308	20.000	27.16
	None-of-the-above		57.933	80.602	67.413
	Homophobia		62.500	83.333	71.429
	Misandry		65.354	47.701	55.15
	Counter-speech		44.681	22.826	30.216
	Misogyny		41.667	48.78	44.944
	Xenophobia		40.000	33.333	36.364
	Transphobic		50.000	50.000	50.000
	Macro/weighted average				
	Macro average		50.555	48.322	47.834
	Weighted average		56.080	64.553	58.529
XLMRoBERTa—Base	Hope-speech	58.286	50.000	24.528	32.911
	None-of-the-above		59.135	85.714	69.986
	Homophobia		75.000	66.667	70.588
	Misandry		67.717	52.121	58.904
	Counter-speech		51.064	24.742	33.333
	Misogyny		50.000	39.344	44.037
	Xenophobia		36.000	33.333	34.615
	Transphobic		0.000	0.000	0.000
	Macro/weighted average				
	Macro average		48.614	40.806	43.047
	Weighted average		58.369	67.714	60.889
XLMRoBERTa—large	Hope-speech	58.083	61.538	27.586	38.095
	None-of-the-above		57.452	85.663	68.777
	Homophobia		62.500	45.455	52.632
	Misandry		70.079	51.149	59.136
	Counter-speech		61.702	26.852	37.419
	Misogyny		39.583	46.341	42.697
	Xenophobia		36.000	36.000	36.000
	Transphobic		0.000	0.000	0.000
	Macro/weighted average				
	Macro average		48.607	39.881	41.844
	Weighted average		58.083	68.096	60.431
MuRIL—base	Hope-speech	59.51	50.000	23.636	32.099
	None-of-the-above		60.817	84.899	70.868
	Homophobia		62.500	71.429	66.667
	Misandry		70.079	48.108	57.051
	Counter-speech		51.064	31.579	39.024
	Misogyny		45.833	42.308	44.000
	Xenophobia		40.000	38.462	39.216
	Transphobic		0.000	0.000	0.000
	Macro/weighted average				
	Macro average		47.537	42.553	43.616
	Weighted average		59.514	67.368	61.547

Table 5 (continued)

Classifiers	Class Labels	Accuracy (%)	Recall (%)	Precision (%)	F1-Score (%)
MuRIL—large	Hope-speech	71.388	80.769	46.667	59.155
	None-of-the-above		74.519	94.225	83.221
	Homophobia		62.500	20.833	31.250
	Misandry		67.717	68.254	67.984
	Counter-speech		63.830	38.462	48.000
	Misogyny		56.250	51.923	54.000
	Xenophobia		76.000	43.182	55.072
	Transphobic		50.000	100.000	66.667
	Macro/weighted average				
	Macro average		66.448	57.943	58.169
	Weighted average		71.388	78.434	73.534
mDeBERTa	Hope-speech	75.393	84.615	51.163	63.768
	None-of-the-above		75.240	95.719	84.253
	Homophobia		87.500	29.167	43.750
	Misandry		77.165	69.504	73.134
	Counter-speech		72.340	50.000	59.130
	Misogyny		62.500	58.824	60.606
	Xenophobia		88.000	50.000	63.768
	Transphobic		0.000	0.000	0.000
	Macro/weighted average				
	Macro average		68.420	50.547	56.051
	Weighted average		75.250	81.020	76.721

evaluate performance. In our study, we used NLPAug for text augmentation and achieved better results, mainly due to the use of adapter-based fine-tuning. Table 6 demonstrates that the adapter-based mDeBERTa achieved a better accuracy compared to other models. However, the models introduced in [69] exhibited stronger performance, as indicated in Table 6. Further, we use Cohen's kappa coefficient which is a statistical measure that evaluates the agreement between two classifiers, considering the potential for chance agreement [75, 76]. It is frequently employed to evaluate performance of the machine learning models in classification tasks. Ranging from -1 to 1, where -1 signifies total disagreement, 0 implies random agreement, and 1 denotes perfect agreement, the kappa coefficient offers insight into the degree of agreement. A higher kappa value signifies superior performance of the model. The Kappa score for the developed models is presented in Table 7. From Table 7, it is understood that some models show substantial agreement (e.g., mDeBERTa, MuRIL—Large, XLM-RoBERTa—Large), others show fair agreement (e.g., mBERT, MuRIL- Base, XLMRoBERTa—Base), indicating varying levels of reliability in their performance for the given classification tasks. To substantiate the efficiency of the proposed models, we conducted a comparative analysis of parameter efficiency between the transformer-based models, elaborated in "Parameter efficiency" section.

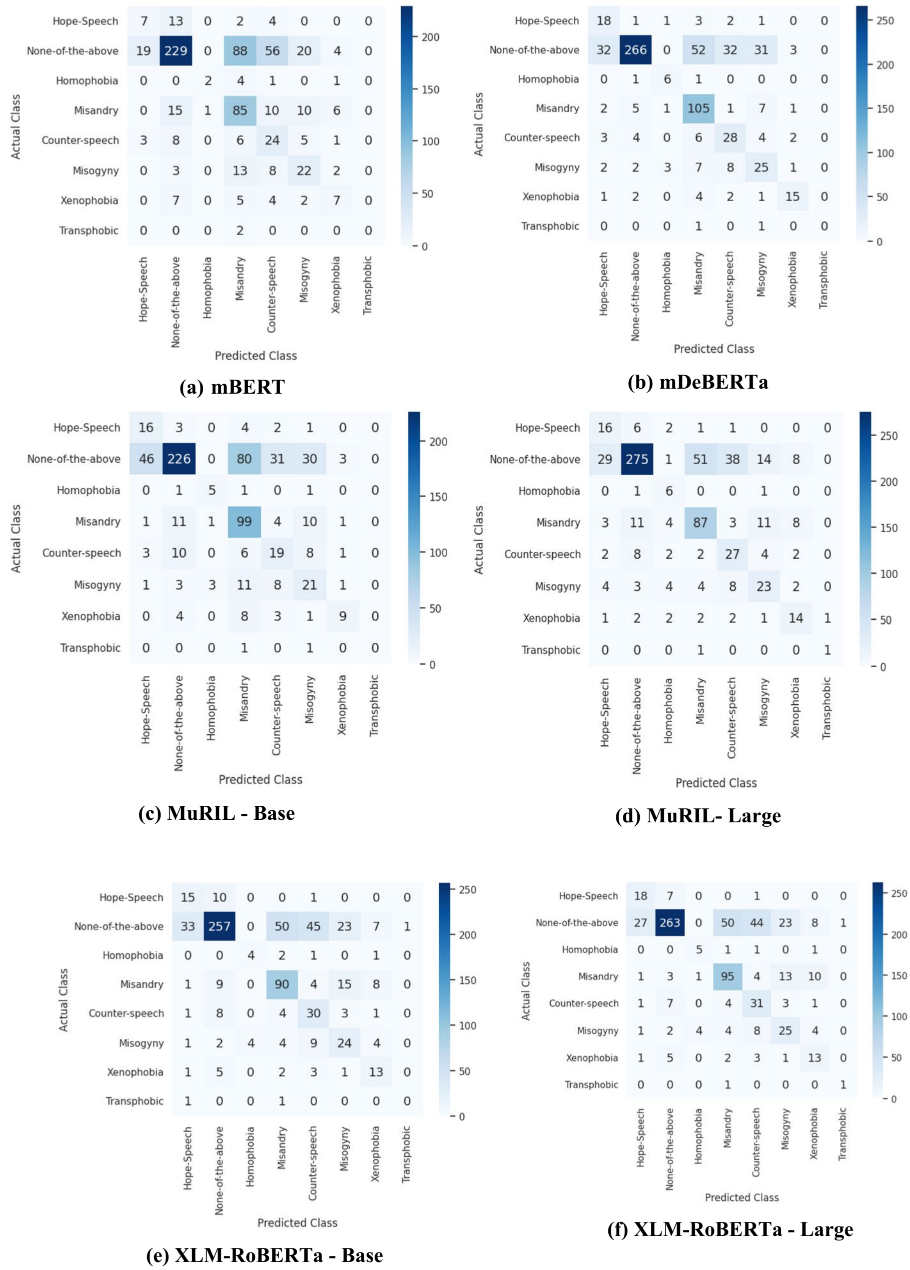

Fig. 2 Confusion matrices for fine-tuned transformer models

Parameter efficiency

The proposed work repurposes pretrained models in three ways: as feature extractors, fine-tuners, and adapters. When used as feature extractors, pretrained model weights are used directly. However, fine-tuning retrained a few top layers, resulting in parameter inefficiency and increased retraining when adapting to new tasks. Adapter modules, on the other hand, as described by Houlsby et al. [18], only need to add a few trainable parameters for each new task, so the whole model doesn't need to be retrained. This approach aims to streamline transfer learning to downstream tasks without extensive

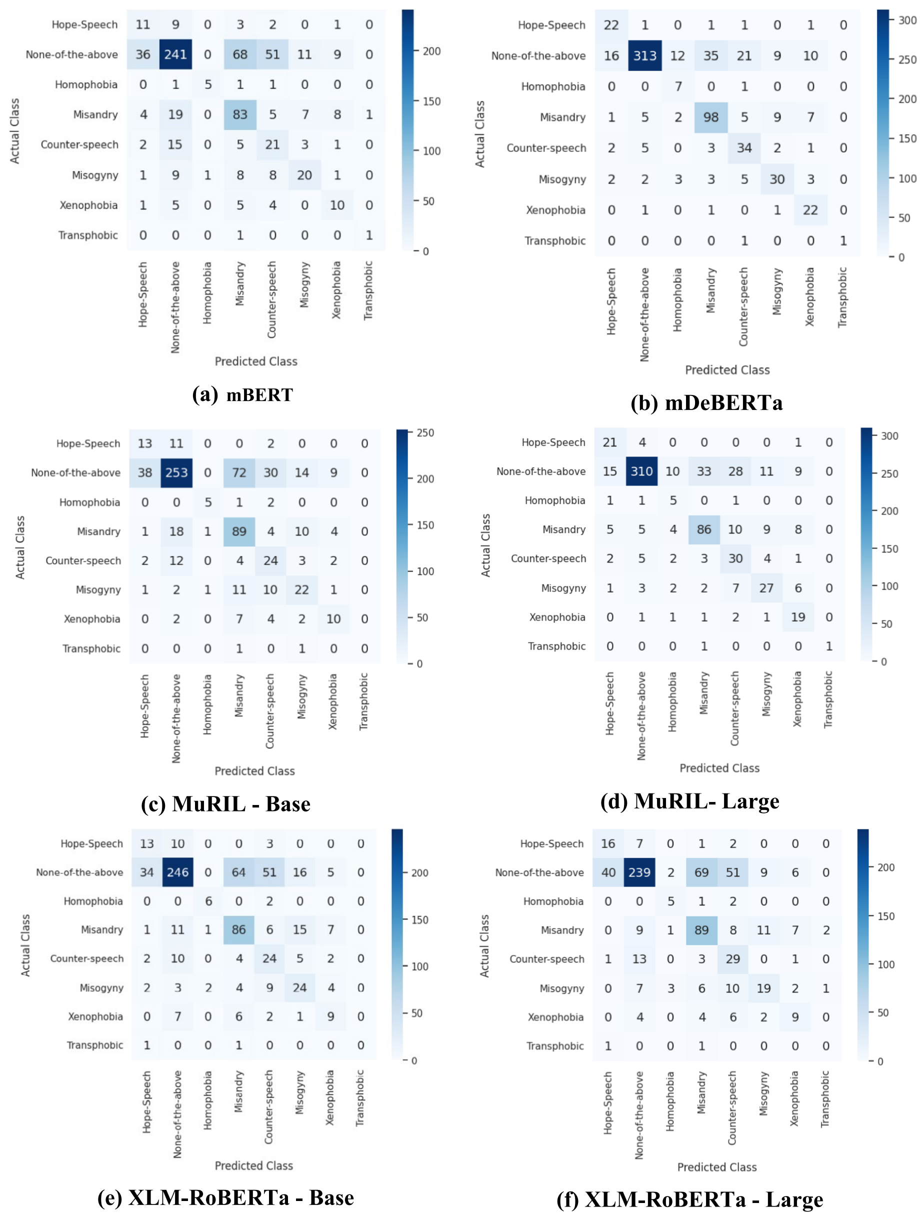

Fig. 3 Confusion matrices for adapter-based transformer models

retraining. Traditional fine-tuning requires retraining the entire model for each new task, increasing the parameter count significantly. Adapter models add modules between pre-trained layers, changing only adapter blocks and normalization weights, which have few parameters. Switching from fine-tuning to adapter-based models improves accuracy and performance. Table 8 lists the trainable parameters for finetuned and adapter-based models.

From Table 8, it is evident that the number of trainable parameters in adapter-based models is significantly fewer when compared to the fine-tuned versions. This implies that adapters are considerably more efficient in terms of time and space requirements than the fine-tuning approach.

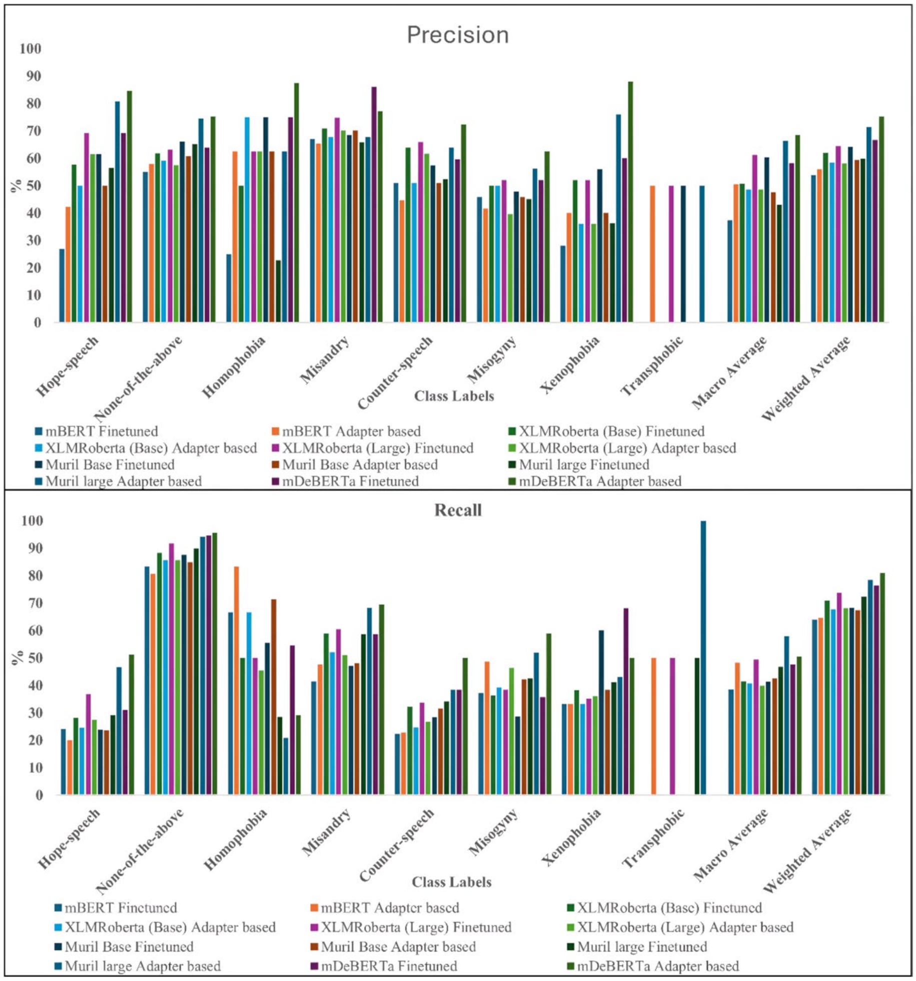

Fig. 4 Finetuned vs Adapter based Transformer models

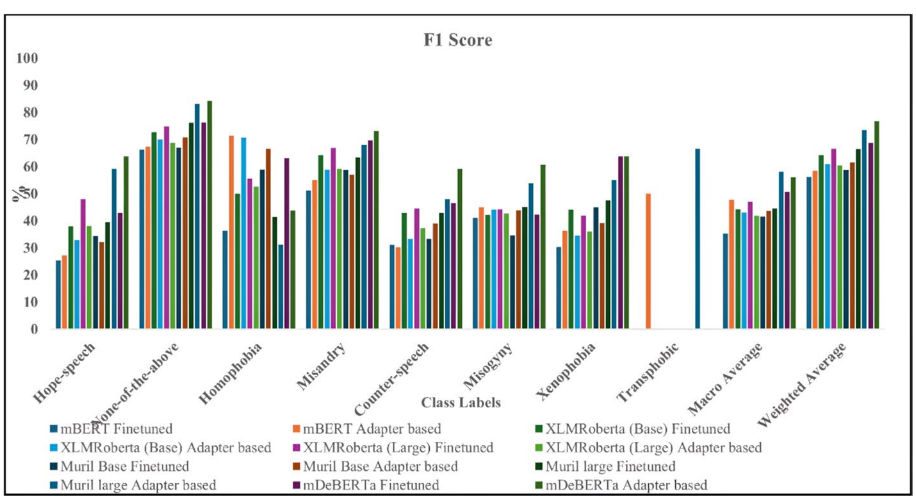

Fig. 4 continued

Table 6 Performance comparison with other models

Reference	Model	Accuracy (%)	Macro Average of (%)		
			Recall	Precision	F1 score
[64]	SVM and TF-IDF with RKS and SMOTE	–	0.38	0.29	0.32
[65]	MuRIL XLM-R-base M-BERT IndicBERT	–	–	–	43 43 40 40
[66]	MuRIL				
	Without augmentation	64	–	–	17
	With augmentation	55	–	–	16
[67]	Indic-BERT	69	22	20	19
[68]	MuRIL	–	–	–	33
[69]	BiLSTM	–	74	67	70
	mBERT	–	64	70	70
[70]	XLMRoBERTa	66	65	66	65
[71]	n-gram-MLP model	–	–	–	12
[72]	BERT	6	–	–	9
[73]	LSTM	46	23	26	22
[74]	Linear SVC	73	38	59	43
Adapter models	mBERT	56.08	50.555	48.322	47.834
	MuRIL—base	59.51	47.537	42.553	43.616
	MuRIL—large	71.388	66.448	57.943	58.169
	XLMRoberta—base	58.286	48.614	40.806	43.047
	XLMRoberta—large	58.083	48.607	39.881	41.844
	mDeBERTa	75.393	68.420	50.547	56.051

Table 7 Cohen's kappa coefficient for the proposed models

Models	Kappa score
Finetuned models	
mBERT	0.3339
mDeBERTa	0.5199
MuRIL—base	0.3854
MuRIL—large	0.4764
XLMRoBERTa—base	0.4503
XLMRoBERTa—large	0.4894
Adapter based models	
mBERT	0.3567
mDeBERTa	0.6318
Muril—base	0.4058
Muril—large	0.5726
XLMRoBERTa—base	0.3993
XLMRoBERTa—large	0.3996

In addition to the adapter-based tuning, there are other parameter-efficient fine-tuning approaches such as BitFit, Prefix-tuning etc. are also available for fine-tuning. We have attempted these two fine-tuning approaches and calculated the number of parameters being retrained. The results are presented in Table 9.

Table 8 Number of parameters retrained

Models	Number of parameters retrained	
	Finetuned	**Adapter-based**
mBERT	177,859,592	1,491,272
MuRIL (base)	237,562,376	1,491,272
MuRIL (large)	505,915,400	4,229,640
XLMRoBERTa (base)	278,049,800	1,491,272
XLMRoBERTa (large)	559,898,632	4,229,640
mDeBERTa	387,923,145	2,156,123

Table 9 Other parameter-efficient fine-tuning approaches

Model	Number of parameters retrained	
	BitFit	**Prefix-tune**
mBERT	150,000	550,000
MuRIL (base)	200,000	800,000
XLM-R (base)	200,000	800,000
XLM-R (large)	500,000	1,750,000
mDeBERTa	350,000	1,250,000

Table 10 Comparison between parameter-efficient finetuning approaches

Feature	BitFit	Prefix-tuning	Adapter-tuning
Changes in base model	No	No	Yes, adapter layers are plugged-in
Interpretability	Simple	Medium	High
Task fit	Shallow tasks	Moderate complexity	High complexity
Language adaptation	Weak	Moderate	Strong
Tamil NLP fit	Limited	Depends on pretraining	Better suited

BitFit fine-tunes only the bias terms and is extremely lightweight but underperforms on complex tasks which require deeper task adaptation. Tamil is morphologically rich and often requires deeper semantic understanding. Hence, BitFit is insufficient to capture complex grammatical patterns in Tamil. Prefix-tuning prepends trainable vectors to the input of each layer which allows task-specific control with minimal updates to the base model. This approach assumes that the models have already learned attention pathways during pretraining. Hence, it is less effective with low-resource or morphologically complex languages (like Tamil) if the base model lacks strong pretraining in those languages. A comparison of these two approaches with adapter-tuning is shown in Table 10.

In this work, we primarily focused on adapter-based tuning as a parameter-efficient fine-tuning strategy due to its proven balance between efficiency and performance in multilingual settings.

Findings and discussion

Six different transformer models were experimented with, examining their effectiveness in this classification task. For fine-tuning, a classification layer was added atop each model, with the top layers retrained for the specific dataset. Instead, adapter-based models added an adapter layer to each transformer layer, and only the weights for the adapter modules needed to be retrained. In previous section of the study, the results have been presented. mDeBERTa is exclusively designed for multilingual tasks and demonstrates strong performance in various NLP tasks across different languages. mDeBERTa is trained on multilingual datasets, allowing it to understand the linguistic structures across various languages. This facilitates effective cross-lingual transfer learning, enabling the model to leverage knowledge learned from one language to understand and process text in another language, such as Tamil. This pretraining helps the model capture the general features of language, which is beneficial when dealing Tamil. Although these models are implicitly trained for multilingual tasks, adapting them to other languages would involve either fine-tuning the model on labeled data from the target language or applying adapter-based methods for low-resource languages. To generalize these models to other languages, the following procedure may be adapted.

- Language-specific fine-tuning: The models can be fine-tuned on a labeled dataset in the target language using the same architecture used for Tamil.
- Adapter-based tuning: For low-resource languages with limited annotated data, lightweight adapter modules can be added, and only those adapters may be trained while keeping the models frozen.
- Cross-lingual transfer: In scenarios where labeled data is scarce, knowledge transfer from a high-resource language to the target language can be performed by using common multilingual representations

Notably, the novelty of this work lies in the integration of adapter modules, enhancing parameter efficiency while maintaining performance. Table 6 further illustrates the effectiveness of the adapter transformer model, highlighting the fact that fine-tuned models require training on a substantially greater number of task-specific parameters than adapter-based transformer models.

While adapter-based transformer models have demonstrated superior performance compared to fine-tuned models, they encountered difficulties when it came to accurately classifying categories within the test data that had a limited number of instances. Notably, when the dataset was tested, the models accurately classified instances that were more prevalent, yet they encountered challenges with categories like Transphobia and Homophobia, resulting in metrics with values of zero. Upon investigating the cause of the reduced accuracy for specific categories, it was identified that dataset imbalance played a significant role. Hence, NLPAug, a Python library for textual augmentation was used and various augmentation techniques, including word-level, contextual word embedding, and random augmentation, were employed. While NLPAug was employed to augment the minority classes, we found that synthetic augmentation often produced repetitive or semantically shallow samples. As a result, the augmented data lacked variety and did not reflect the true nature of such comments. Despite augmenting the training dataset by replacing synonyms and contextual words, the study suggests that further performance improvements could be attained by incorporating actual sentences for the infrequent classes rather than simple augmentation. A notable observation was that most models struggled with classifications in the Transphobia class. Even with the augmentation of infrequent classes, it has been found that the generated texts did not fully represent the Transphobia class, being only shuffled or rephrased comments. Consequently, this led to a substantial number of misclassifications within the test dataset.

Further manual examination of misclassified samples in the Transphobia class revealed that the test instances are not linguistically clear, and included indirect references that did not appear in the training set. This domain-specific linguistic complexity, coupled with the extremely low support (only 2 instances in the test set), complicated the ability of model to generalize effectively. For instance, the comment "எச் ராஜா பெண்" ("H. Raja Girl") is tagged as Transphobic in the dataset. While this comment may seem neutral in isolation, it functions as an insult, implying that the male is being feminized disrespectfully. Due to the absence of explicit offensive language, most models (except XLM-RoBERTa-Large, MuRIL-Large-Adapter, and mBERT-Adapter) misclassified this as None-of-the-above, highlighting their inability to interpret the embedded abuse. This issue may be addressed by (1) manual curation or crowdsourcing of diverse, real-world examples from social media platforms where Transphobic or Homophobic speech is prevalent, (2) using large language models like GPT-4 to generate synthetic but realistic and diverse abusive content, (3) using class-weighted cross-entropy or focal loss and (4) using few-shot or contrastive learning approaches for better representation of low-resource classes.

Further, the developed models exhibit high class level accuracy for almost all the classes, but poor performance against precision, recall and F1-score. For instance, in case of mBERT (feature extractor), the class level accuracy is 94.13% for class 1 (Hope Speech), whereas other metrics are low. The high accuracy is driven by the large number of true negatives (652 out of 699 samples). This is because class level accuracy considers both true positives and true negatives. When the dataset is imbalanced, the large number of true negatives can inflate the class-level accuracy. Since there are few true positives (6), relatively more false positives (21) and false negatives (20), the precision, recall, and F1-score are low.

In this work, we focused on addressing the severe class imbalance in the dataset, as shown in Table 1, through oversampling and under-sampling-based text augmentation, which allowed us to train the models on a more balanced dataset. As the original dataset had highly skewed class distributions, for instance only 6 instances for the Transphobic class, we directly proceeded with text augmentation for training. We did not perform experiments on the original unaugmented data, as the imbalance significantly would affect model learning performance. We plan to perform a comparative analysis to quantify the real impact of augmentation on classifier performance and generalization.

Furthermore, we have also assessed the training time, memory consumption, and inference delay of the models to provide a better picture of the computing effects of multilingual transformer models and presented in Table 11.

With the lowest inference delay of 31 ms per comment, 7.1 min per epoch, and 6 GB of GPU memory, mBERT was the most computationally efficient model among the others.

Table 11 Computational cost and inference latency

Model	Training time per epoch (3742 samples, batch size 16) (app.) (in min)	Memory usage (app.) (in GBs)	Inference latency per comment (app.) (in ms)
mBERT	7.1	6	31
MuRIL (base)	8.2	7	36
MuRIL (large)	15.9	13	63
XLM-RoBERTa (base)	9.6	7.5	38
XLM-RoBERTa (large)	18.2	15.2	69
mDeBERTa	12	11	54

On the other hand, larger models like XLM-RoBERTa-large and MuRIL-large showed noticeably higher computing needs, with inference latencies surpassing 60 ms, memory utilization of 13 GB and 15.2 GB, and training times of 15.9 and 18.2 min. With a moderate computational cost and latency, mDeBERTa demonstrated a balanced trade-off, making it a good choice for situations requiring both linguistic depth and efficiency. This investigation highlights how crucial it is to choose models based on computational viability as well as classification performance, particularly when deploying in contexts with limited resources.

Statistical analysis A t-test helps determine if the observed difference in performance between two models is due to actual performance differences or just random chance. A t-test on accuracy, macro precision, macro recall, and macro F1 score was performed to assess the statistical significance of the performance of the models and determine whether they were the result of pure chance. We compared each model to the best-performing model using the above metrics. Further, t-tests provide two significant metrics namely: T-Statistic and P-Value. The difference between the averages of two groups and the variation within the groups is indicated by the T-statistic. It indicates the standard error difference between the group averages. If the null hypothesis, that is, there is no actual difference between the groups—is true, the P-value indicates the probability of obtaining the observed results. According to standard statistical interpretation: (i) A p-value less than 0.05 (i.e., $p < 0.05$) is considered statistically significant, indicting that we can reject the null hypothesis and conclude that the performance difference between the models is not due to chance. (ii) A p-value greater than 0.05 indicates that the difference in performance is not statistically significant and could be due to random chance. The results of the t-tests are presented from Tables 12, 13, 14, 15.

In Tables 12, 13, 14, 15, a negative t-statistic suggests that the first model performed worse than the second model in comparison. For instance, the t-statistic of − 15.77 when comparing mBERT and MuRIL (Large) indicates that performance of mBERT is significantly lower than that of MuRIL (Large). Conversely, a positive t-statistic indicates the first model performed better than the second (notably seen in comparisons involving MuRIL (Large) and XLMRoBERTa scores). Because the p-value is less than the significance level often 0.05, the null hypothesis is rejected which indicates that there is notable difference in performance of the models. In Tables 12, 13, 14, 15, the results show that the comparison of all the models with mDeBERTa (e.g., mBERT, MuRIL-large, XLM-RoBERTa-base) yield p-values below 0.05, indicating that the performance differences

Table 12 T-statistics for accuracy

Model 1	Model 2	T-statistic	P-value	Significant difference (p < 0.05)
mBERT	MuRIL (base)	− 11.58	0.0014	Yes
mBERT	MuRIL (large)	− 23.51	0.00017	Yes
mBERT	XLMRoBERTa (BASE)	− 3.86	0.0307	Yes
mBERT	XLMRoBERTa (large)	− 2.71	0.0732	No
mBERT	mDeBERTa	− 26.9	0.00011	Yes
MuRIL (base)	MuRIL (large)	− 27.21	0.00011	Yes
MuRIL (base)	XLMRoBERTa (base)	2.88	0.0636	No
MuRIL (base)	XLMRoBERTa (large)	1.15	0.3345	No
MuRIL (base)	mDeBERTa	− 33.4	5.90E−05	Yes
MuRIL (large)	XLMRoBERTa (base)	81.15	4.12E−06	Yes
MuRIL (large)	XLMRoBERTa (large)	29.65	8.43E−05	Yes
MuRIL (large)	mDeBERTa	− 44.06	2.57E−05	Yes
XLMRoBERTa (base)	XLMRoBERTa (Large)	− 0.65	0.5625	No
XLMRoBERTa (base)	mDeBERTa	− 136.56	8.66E−07	Yes
XLMRoBERTa (large)	mDeBERTa	− 48.67	1.91E−05	Yes

Table 13 T-statistics for macro precision

Model 1	Model 2	T-statistic	P-value	Significant difference (p < 0.05)
mBERT	MuRIL (base)	10.44	0.0019	Yes
mBERT	MuRIL (large)	− 15.77	0.00055	Yes
mBERT	XLMRoBERTa (base)	14.25	0.00075	Yes
mBERT	XLMRoBERTa (Large)	15.34	0.0006	Yes
mBERT	mDeBERTa	− 3.55	0.0382	Yes
MuRIL (base)	MuRIL (large)	− 180.34	3.76E−07	Yes
MuRIL (base)	XLMRoBERTa (base)	62.68	8.95E−06	Yes
MuRIL (base)	XLMRoBERTa (Large)	9.65	0.0024	Yes
MuRIL (base)	mDeBERTa	− 19.62	0.00029	Yes
MuRIL (large)	XLMRoBERTa (base)	163.54	5.04E−07	Yes
MuRIL (large)	XLMRoBERTa (large)	66.78	7.40E−06	Yes
MuRIL (large)	mDeBERTa	15.25	0.00061	Yes
XLMRoBERTa (base)	XLMRoBERTa (large)	3.04	0.056	No
XLMRoBERTa (base)	mDeBERTa	− 24.58	0.00015	Yes
XLMRoBERTa (large)	mDeBERTa	− 16.02	0.00053	Yes

are statistically significant. However, in a few cases (e.g., XLM-RoBERTa-base vs. XLM-RoBERTa-large, p = 0.5625), the differences are not statistically significant, indicating those models performed similarly in terms of accuracy. These findings reiterate the conclusion that the performance of mDeBERTa is not only numerically better but also statistically meaningful compared to other models.

Implications for real-world deployment and scalability In addition to achieving good performance, our experiments demonstrate that adapter-based transformer models,

Table 14 T-statistics for macro recall

Model 1	Model 2	T-statistic	P-value	Significant difference (p < 0.05)
mBERT	MuRIL (Base)	38.47	3.86E−05	Yes
mBERT	MuRIL (large)	− 98.87	2.28E−06	Yes
mBERT	XLMRoBERTa (base)	81.04	4.14E−06	Yes
mBERT	XLMRoBERTa (large)	4.92	0.0161	Yes
mBERT	mDeBERTa	− 385.74	3.84E−08	Yes
MuRIL (base)	MuRIL (large)	− 191.29	3.15E−07	Yes
MuRIL (base)	XLMRoBERTa (base)	− 9.34	0.0026	Yes
MuRIL (base)	XLMRoBERTa (large)	− 1.86	0.1602	No
MuRIL (base)	mDeBERTa	− 205.36	2.55E−07	Yes
MuRIL (large)	XLMRoBERTa (base)	102.4	2.05E−06	Yes
MuRIL (large)	XLMRoBERTa (large)	37.8	4.07E−05	Yes
MuRIL (large)	mDeBERTa	− 8.65	0.0032	Yes
XLMRoBERTa (base)	XLMRoBERTa (large)	0.22	0.8368	No
XLMRoBERTa (base)	mDeBERTa	− 377.1	4.11E−08	Yes
XLMRoBERTa (large)	mDeBERTa	− 50.23	1.74E−05	Yes

Table 15 T-statistics for macro F1-score

Model 1	Model 2	T-statistic	P-value	Significant difference (p < 0.05)
mBERT	MuRIL (base)	16.38	0.0005	Yes
mBERT	MuRIL (large)	− 39.73	3.51E−05	Yes
mBERT	XLMRoBERTa (base)	18.41	0.00035	Yes
mBERT	XLMRoBERTa (large)	40.94	3.21E−05	Yes
mBERT	mDeBERTa	− 23.32	0.00017	Yes
MuRIL (base)	MuRIL (large)	− 235.29	1.69E−07	Yes
MuRIL (base)	XLMRoBERTa (base)	16.8	0.00046	Yes
MuRIL (base)	XLMRoBERTa (large)	5.42	0.012	Yes
MuRIL (base)	mDeBERTa	− 44.86	2.44E− 05	Yes
MuRIL (large)	XLMRoBERTa (base)	193.87	3.03E− 07	Yes
MuRIL (large)	XLMRoBERTa (large)	79	4.47E−06	Yes
MuRIL (large)	mDeBERTa	10.19	0.002	Yes
XLMRoBERTa (base)	XLMRoBERTa (large)	2.34	0.101	No
XLMRoBERTa (base)	mDeBERTa	− 50.59	1.70E−05	Yes
XLMRoBERTa (large)	mDeBERTa	− 33.62	5.78E−05	Yes

such as mDeBERTa, are parameter-efficient, requiring fewer retrainable parameters than fine-tuned models. In low-resource or latency-sensitive contexts, this makes mobile moderation tools (Mobile moderation tools are the apps that help detect or filter harmful content), browser plugins, or on-device AI systems more scalable for real-world deployment. However, even after using text augmentation, our results clearly show the challenges in identifying abuse in rare or infrequent classes (such as transphobia). This raises an important issue that the models trained on unbalanced datasets might not identify harmful or offensive contents, resulting in false negatives, which is a notable issue in applications that depend online safety.

In terms of scalability, base models with adapter provide a balance between performance and resource utilization. But the large transformer models such as XLM-R Large, MuRIL Large, while accurate in some scenarios, present issues due to large memory and inference latency, restricting usage unless enabled by cloud or server-based backends. Furthermore, reinforcement learning, continuous learning, incremental learning, and other techniques will enable models to dynamically adapt to the changing nature of language and slang usage. These results indicate that even though transformer-based models have potential for identifying social media abuse, scalable implementation necessitates careful evaluation of model size, class balance, and contextual awareness.

Although the objective of our research is to develop models for identifying abusive content in Tamil social media texts, we believe that these systems might be misused, especially in situations where moderation may be abused or used to prevent free speech. Over-censorship is one potential abuse, where the models mistakenly label comments that are not abusive or critical—such as comedy, political protest, or minority voices—as abusive. If the model has trained from biased or unbalanced data, this may lead to bias against communities, the removal of unfair contents or the silence of oppressed groups.

Error analysis

Since much of the test dataset contains "None of the above" class instances, the metric wise performance of this class is found to be better than other classes. The class "None of the above" is used to label comments that do not fall into any of the other seven classes in the dataset. It is not indicative of a classification error but rather a specific label. For instance, the comment "சக தோழியாக நினைத்து ஆதரவு தாருங்கள் நண்பர்களே!!!! பயனுள்ள வகையில் தகவல்களை பதிவிடுகிறேனன் பார்த்து விட்டு ஆதரவு தாருங்கள் (Friends, Support as colleague!!!! I am posting information in a useful manner so please support me), actually belongs to "None of the above" class and all the models have correctly classified. But, the comment, சோறு எவ்வளவு முக்கியமோ அதே மாதிரி சாதி முக்கியம் (Caste is as important as food) is classified as "None of the above", but the models mBERT and MuRIL (Base and Large) predicted it as "Counter Speech". These models might have interpreted the sentence "Caste is as important as food" in a way that fits with what we usually associate with counter-speech and could have interpreted that the sentence involved questions about social order and how people are classified based on their caste. It means it emphasizes equality, fairness, and equal opportunities for all. But the other models such as mDBeRTa and XLMRoBERTa have predicted the sentence correctly. These models have understood it as a comparison between the significance of caste and the necessity of food, without any explicit abusive connotations.

Conclusion and future work

This study examined the efficiency of various pre-trained multilingual transformer models, including mBERT, MuRIL (base and big), XLM-RoBERTa (base and large), and mDeBERTa, for classifying abusive text in Tamil. Three methods were used in the adaptation of these models: adapter-based tuning, fine-tuning, and feature extraction. Among these, the adapter-based mDeBERTa model showed promise for adapter-tuning for the tasks in low-resource languages by performing well while requiring a smaller number of trainable parameters when compared to other models. In addition, this study offers a few insights. First, adapter-based approaches strike a balance between efficiency and effectiveness, making them suitable for deployment in resource-constrained environments. Second, despite text augmentation efforts using tools like NLPAug, the models struggled with minority classes such as Transphobic and Homophobic, indicating that synthetic augmentation alone may not sufficiently capture the linguistic meanings and diversity present in real-world abusive content. The broader implication of this research lies in demonstrating that parameter-efficient multilingual models can be both scalable and impactful in regional language contexts. As online content moderation becomes increasingly necessary in diverse linguistic settings, such models can support inclusive and equitable digital environment for underrepresented communities.

We intend to investigate more sophisticated data augmentation methods in the future, such as task-specific pretraining on abusive speech corpora and LLM-assisted synthesis of minority class examples. To enhance performance on unbalanced datasets, we also plan to use cost-sensitive learning strategies including class-balanced sampling and focused loss. By using cross-lingual transfer learning to improve generalization and robustness across closely related languages, we plan to expand this research to Dravidian languages other than Tamil. Additionally, to handle low-data regimes that are typical in abusive content detection tasks, we will investigate few-shot and meta-learning techniques.

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Author contributions
M.S., S.V.E., N.S., and U.M. contributed to the study design and manuscript preparation. M.S. and N.S. conducted the experiments and analyzed the data. S.V.E. and U.M. assisted in data interpretation and manuscript revision. All authors reviewed and approved the final manuscript.

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Data availability
No datasets were generated or analysed during the current study.

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References

1. Butt S et al. Sexism identification using BERT and data augmentation-EXIST2021. In: IberLEF@ SEPLN; 2021.
2. Spertus E. Smokey: automatic recognition of hostile messages. In: AAAI/IAAI; 1997.
3. Razavi AH et al. Offensive language detection using multi-level classification. In: Advances in Artificial Intelligence: 23rd Canadian conference on Artificial Intelligence, Canadian AI 2010, Ottawa, Canada, May 31–June 2, 2010. Proceedings 23. Springer; 2010.

4. Djuric N et al. Hate speech detection with comment embeddings. In: Proceedings of the 24th international conference on world wide web; 2015.
5. Chatzakou D et al. Mean birds: Detecting aggression and bullying on twitter. In: Proceedings of the 2017 ACM on web science conference; 2017.
6. Chakravarthi BR et al. Overview of the track on HASOC-offensive language identification-DravidianCodeMix. in FIRE (Working notes); 2020.
7. Suryawanshi S, Chakravarthi BR. Findings of the shared task on troll meme classification in Tamil. In: Proceedings of the first workshop on speech and language technologies for Dravidian languages; 2021.
8. Amjad M et al. Urduthreat@ fire2021: shared track on abusive threat identification in Urdu. In: Proceedings of the 13th annual meeting of the forum for information retrieval evaluation; 2021.
9. Shared task on abusive comment detection; 2022.
10. Devlin J et al. Bert: pre-training of deep bidirectional transformers for language understanding; 2018. arXiv preprint arXiv:1810.04805.
11. He P et al. Deberta: decoding-enhanced bert with disentangled attention; 2020. arXiv preprint arXiv:2006.03654.
12. Peters ME, Ruder S, Smith NA. To tune or not to tune? Adapting pretrained representations to diverse tasks; 2019. arXiv preprint arXiv:1903.05987.
13. Conneau A et al. Unsupervised cross-lingual representation learning at scale; 2019. arXiv preprint arXiv:1911.02116.
14. Dave B, Bhat S, Majumder P. IRNLP_DAIICT@ DravidianLangTech-EACL2021: offensive language identification in Dravidian languages using TF-IDF char n-grams and MuRIL. In: Proceedings of the first workshop on speech and language technologies for Dravidian Languages; 2021.
15. Chakravarthi BR et al. Corpus creation for sentiment analysis in code-mixed Tamil-English text; 2020. arXiv preprint arXiv:2006.00206.
16. Mahabadi RK et al. Parameter-efficient multi-task fine-tuning for transformers via shared hypernetworks; 2021. arXiv preprint arXiv:2106.04489.
17. Semnani S, Sadagopan KR, Tlili F. BERT-A: Finetuning BERT with adapters and data augmentation. Standford University. 2019;1:2790–9.
18. Houlsby N et al. Parameter-efficient transfer learning for NLP. In: International conference on machine learning; 2019. PMLR.
19. He R et al. On the effectiveness of adapter-based tuning for pretrained language model adaptation; 2021. arXiv preprint arXiv:2106.03164.
20. Chakravarthi BR et al. Findings of the shared task on offensive language identification in Tamil, Malayalam, and Kannada. In: Proceedings of the first workshop on speech and language technologies for Dravidian languages; 2021.
21. "Abusive comment detection in Tamil" 2022. https://competitions.codalab.org/competitions/36403.
22. Priyadharshini R et al. Overview of abusive comment detection in Tamil-ACL 2022. In: Proceedings of the second workshop on speech and language technologies for Dravidian Languages; 2022.
23. Hegde A et al. Overview of the shared task on machine translation in Dravidian languages. In: Proceedings of the second workshop on speech and language technologies for Dravidian languages; 2022.
24. "Homophobia/transphobia detection in social media comments,"; 2022. https://competitions.codalab.org/competitions/36394
25. "Hate speech and offensive content identification,"; 2021. https://hasocfire.github.io/hasoc/2021/index.html.
26. Modha S et al. Overview of the hasoc subtrack at fire 2021: hate speech and offensive content identification in english and indo-aryan languages and conversational hate speech. In: Proceedings of the 13th annual meeting of the forum for information retrieval evaluation; 2021.
27. Ashraf N, Zubiaga A, Gelbukh A. Abusive language detection in youtube comments leveraging replies as conversational context. PeerJ Comput Sci. 2021;7: e742.
28. Lee Y, Yoon S, Jung K. Comparative studies of detecting abusive language on twitter; 2018. arXiv preprint arXiv:1808.10245.
29. Emon EA et al. A deep learning approach to detect abusive Bengali text. InL 2019 7th international conference on smart computing & communications (ICSCC). IEEE; 2019.
30. Aurpa TT, Sadik R, Ahmed MS. Abusive Bangla comments detection on Facebook using transformer-based deep learning models. Soc Netw Anal Min. 2022;12(1): 24.
31. Sharif O, Hossain E, Hoque MM. Nlp-cuet@ dravidianlangtech-eacl2021: offensive language detection from multilingual code-mixed text using transformers; 2021. arXiv preprint arXiv:2103.00455.
32. Pitsilis GK, Ramampiaro H, Langseth H. Detecting offensive language in tweets using deep learning; 2018. arXiv preprint arXiv:1801.04433.
33. Ziehe S, Pannach F, Krishnan A. GCDH@ LT-EDI-EACL2021: XLM-RoBERTa for hope speech detection in English, Malayalam, and Tamil. In: proceedings of the first workshop on language Technology for Equality, diversity and inclusion; 2021.
34. Glazkova A, Kadantsev M, Glazkov M. Fine-tuning of pre-trained transformers for hate, offensive, and profane content detection in English and Marathi; 2021. arXiv preprint arXiv:2110.12687.
35. Steimel K et al. Investigating multilingual abusive language detection: a cautionary tale. In: Proceedings of the international conference on recent advances in natural language processing (RANLP 2019); 2019.
36. El-Alami F-Z, El Alaoui SO, Nahnahi NE. A multilingual offensive language detection method based on transfer learning from transformer fine-tuning model. J King Saud Univ Comput Inf Sci. 2022;34(8):6048–56.
37. Sundar A, et al. Hope speech detection for dravidian languages using cross-lingual embeddings with stacked encoder architecture. SN Comput Sci. 2022;3:1–15.
38. Chakravarthi BR. HopeEDI: a multilingual hope speech detection dataset for equality, diversity, and inclusion. In: Proceedings of the third workshop on computational modeling of People's opinions, personality, and emotion's in Social Media; 2020.

39. Chakravarthi BR et al. Dataset for identification of homophobia and transophobia in multilingual YouTube comments; 2021. arXiv preprint arXiv:2109.00227.
40. Jose N et al. A survey of current datasets for code-switching research. In: 2020 6th international conference on advanced computing and communication systems (ICACCS). IEEE; 2020.
41. Sharma D, Gupta V, Singh VK. Abusive comment detection in Tamil using deep learning. In: Computational intelligence methods for sentiment analysis in natural language processing applications. Elsevier; 2024. p. 207–26.
42. Rehman MZU, et al. User-aware multilingual abusive content detection in social media. Inf Process Manage. 2023;60(5): 103450.
43. Chakravarthi BR, et al. Detecting abusive comments at a fine-grained level in a low-resource language. Nat Lang Process J. 2023;3: 100006.
44. Sultana S et al. Detection of abusive Bengali comments for mixed social media data using machine learning; 2023.
45. Ibrohim MO, Budi I. Hate speech and abusive language detection in Indonesian social media: progress and challenges. Heliyon. 2023. https://doi.org/10.1016/j.heliyon.2023.e18647.
46. Khan A, et al. Abusive language detection in Urdu text: leveraging deep learning and attention mechanism. IEEE Access. 2024. https://doi.org/10.1109/ACCESS.2024.3370232.
47. Devi VS, Kannimuthu S, Madasamy AK. The effect of phrase vector embedding in explainable hierarchical attention-based Tamil code-mixed hate speech and intent detection. IEEE Access. 2024. https://doi.org/10.1109/ACCESS.2024.3349958.
48. Vinoth D, Prabhavathy P. Automated sarcasm detection and classification using hyperparameter tuned deep learning model for social networks. Expert Syst. 2022;39(10): e13107.
49. Osmani A, Mohasefi JB, Gharehchopogh FS. Enriched latent dirichlet allocation for sentiment analysis. Expert Syst. 2020;37(4): e12527.
50. Asghar MZ, et al. Creating sentiment lexicon for sentiment analysis in Urdu: the case of a resource-poor language. Expert Syst. 2019;36(3): e12397.
51. Vaswani A et al. Attention is all you need. Advances in neural information processing systems; 2017. p. 30.
52. Alammar J. The illustrated transformer. The illustrated transformer–Jay Alammar–visualizing machine learning one concept at a time.
53. Lan Z et al. Albert: a lite bert for self-supervised learning of language representations; 2019. arXiv preprint arXiv:1909.11942.
54. Pires, T., E. Schlinger, and D. Garrette, How multilingual is multilingual BERT? arXiv preprint arXiv:1906.01502, 2019.
55. Liu Y et al. Roberta: a robustly optimized bert pretraining approach; 2019. arXiv preprint arXiv:1907.11692.
56. Aßenmacher M, Heumann C. On the comparability of pre-trained language models; 2020. arXiv preprint arXiv:2001.00781.
57. Hu J et al. Xtreme: a massively multilingual multi-task benchmark for evaluating cross-lingual generalisation. In: International conference on machine learning; 2020. PMLR.
58. Howard J, Ruder S. Universal language model fine-tuning for text classification; 2018. arXiv preprint arXiv:1801.06146.
59. Pfeiffer J et al. AdapterFusion: non-destructive task composition for transfer learning; 2020. arXiv preprint arXiv:2005.00247.
60. Pfeiffer J et al. Adapterhub: a framework for adapting transformers; 2020. arXiv preprint arXiv:2007.07779.
61. Kim S et al. Revisiting pretraining with adapters. In Proceedings of the 6th workshop on representation learning for NLP (RepL4NLP-2021); 2021.
62. "Hugging Face." https://huggingface.co/docs/transformers/index.
63. Priyadharshini R et al. Findings of the shared task on Abusive Comment Detection in Tamil. In: Proceedings of the second workshop on speech and language technologies for Dravidian languages. Association for Computational Linguistics; 2022.
64. Prasanth S et al. CEN-Tamil@ DravidianLangTech-ACL2022: abusive comment detection in Tamil using TF-IDF and random kitchen sink algorithm. In: Proceedings of the second workshop on speech and language technologies for Dravidian languages; 2022.
65. Patankar S et al. Optimize_Prime@ DravidianLangTech-ACL2022: abusive comment detection in Tamil; 2022. arXiv preprint arXiv:2204.09675.
66. Pahwa B. BpHigh@ TamilNLP-ACL2022: effects of data augmentation on indic-transformer based classifier for abusive comments detection in Tamil. In: Proceedings of the second workshop on speech and language technologies for Dravidian languages; 2022.
67. Hossain A et al. COMBATANT@ TamilNLP-ACL2022: fine-grained categorization of abusive comments using logistic regression. In: Proceedings of the second workshop on speech and language technologies for Dravidian languages; 2022.
68. Palanikumar V et al. DE-ABUSE@ TamilNLP-ACL 2022: transliteration as data augmentation for abuse detection in Tamil. In: Proceedings of the second workshop on speech and language technologies for Dravidian languages; 2022.
69. Rajalakshmi R, Duraphe A, Shibani A. DLRG@ DravidianLangTech-ACL2022: abusive comment detection in Tamil using multilingual transformer models. In: Proceedings of the second workshop on speech and language technologies for Dravidian languages; 2022.
70. Prasad G, Prasad J, Gunavathi C. GJG@ TamilNLP-ACL2022: using transformers for abusive comment classification in Tamil. In: Proceedings of the second workshop on speech and language technologies for Dravidian languages; 2022.
71. Balouchzahi F et al. MUCIC@ TamilNLP-ACL2022: abusive comment detection in Tamil language using 1D Conv-LSTM. In: Proceedings of the second workshop on speech and language technologies for Dravidian languages; 2022.

72. Bharathi B, Varsha J. SSNCSE NLP@ TamilNLP-ACL2022: Transformer based approach for detection of abusive comment for Tamil language. In: Proceedings of the second workshop on speech and language technologies for Dravidian languages; 2022.
73. Yigezu MG et al. Habesha@ dravidianlangtech: abusive comment detection using deep learning approach. In: Proceedings of the third workshop on speech and language technologies for Dravidian languages; 2023.
74. Reddy A, Moorthi P, Thomas AM. Supernova@ dravidianlangtech 2023@ abusive comment detection in Tamil and Telugu-(Tamil, Tamil-English, Telugu-English). In: Proceedings of the third workshop on speech and language technologies for Dravidian languages; 2023.
75. Khan S, Qasim I, Khan W, Aurangzeb K, Khan JA, Anwar MS. A novel transformer attention-based approach for sarcasm detection. Expert Syst. 2025;42(1): e13686.
76. Khan S, Qasim I, Khan W, Khan A, Ali Khan J, Qahmash A, Ghadi YY. An automated approach to identify sarcasm in low-resource language. PLoS ONE. 2024;19(12): e0307186.